Minimização de Tabela de Estados

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Objetivo – encontrar uma tabela com menos estados mas com o mesmo comportamento.

Como? – Verificando se existem estados equivalentes. Se dois estados são equivalentes posso excluir um deles da tabela sem alterar o funcionamento do autômato.

Condição para que dois estados sejam equivalentes:

- Saídas iguais para todas as entradas,

- Próximos estados equivalentes para todas as entradas

Minimize a tabela de estados abaixo

	x	
q	0	1
1	2,0	3,0
2	4,0	5,0
3	6,0	7,0
4	8,0	9,0
5	10,0	11,0
6	4,0	12,0
7	10,0	12,0
8	8,0	1,0
9	10,1	1,0
10	4,0	1,0
11	2,0	1,0
12	2,0	1,0

1º passo – Inspeção da tabela

Verificar se existem estados idênticos

	x	
q	0	1
1	2,0	3,0
2	4,0	5,0
3	6,0	7,0
4	8,0	9,0
5	10,0	11,0
6	4,0	12,0
7	10,0	12,0
8	8,0	1,0
9	10,1	1,0
10	4,0	1,0
11	2,0	1,0
~~12~~	~~2,0~~	~~1,0~~

Estado 12 = Estado 11

Elimino o estado 12 e substituo por 11.

	x	
q	0	1
1	2,0	3,0
2	4,0	5,0
3	6,0	7,0
4	8,0	9,0
5	10,0	11,0
6	4,0	11,0
7	10,0	11,0
8	8,0	1,0
9	10,1	1,0
10	4,0	1,0
11	2,0	1,0

	x	
q	0	1
1	2,0	3,0
2	4,0	5,0
3	6,0	7,0
4	8,0	9,0
5	10,0	11,0
6	4,0	11,0
~~7~~	~~10,0~~	~~11,0~~
8	8,0	1,0
9	10,1	1,0
10	4,0	1,0
11	2,0	1,0

Estado 7 = Estado 5

Elimino o estado 7 e substituo por 5.

	x	
q	0	1
1	2,0	3,0
2	4,0	5,0
3	6,0	5,0
4	8,0	9,0
5	10,0	11,0
6	4,0	11,0
8	8,0	1,0
9	10,1	1,0
10	4,0	1,0
11	2,0	1,0

Tabela de Implicação

	1	2	3	4	5	6	8	9	10
2									
3									
4									
5									
6									
8									
9									
10									
11									

	x	
q	0	1
1	2,0	3,0
2	4,0	5,0
3	6,0	5,0
4	8,0	9,0
5	10,0	11,0
6	4,0	11,0
8	8,0	1,0
9	10,1	1,0
10	4,0	1,0
11	2,0	1,0

Tabela de Implicação

	1	2	3	4	5	6	8	9	10
2									
3									
4									
5									
6									
8									
9	X	X	X	X	X	X	X		
10								X	
11								X	

	x	
q	0	1
1	2,0	3,0
2	4,0	5,0
3	6,0	5,0
4	8,0	9,0
5	10,0	11,0
6	4,0	11,0
8	8,0	1,0
9	10,1	1,0
10	4,0	1,0
11	2,0	1,0

Tabela de Implicação

	1	2	3	4	5	6	8	9	10
2	2 - 4 3 - 5								
3	2 - 6 3 - 5								
4	2 - 8 3 - 9								
5	2 - 10 3 - 11								
6	2 - 4 3 - 11								
8	2 - 8 1 - 3								
9	X	X	X	X	X	X	X		
10	2 - 4 1 - 3							X	
11	1 - 3							X	

	x	
q	0	1
1	2,0	3,0
2	4,0	5,0
3	6,0	5,0
4	8,0	9,0
5	10,0	11,0
6	4,0	11,0
8	8,0	1,0
9	10,1	1,0
10	4,0	1,0
11	2,0	1,0

Tabela de Implicação

	1	2	3	4	5	6	8	9	10
2	2 - 4 3 - 5								
3	2 - 6 3 - 5	4 - 6							
4	2 - 8 3 - 9	4 - 8 5 - 9	6 - 8 5 - 9						
5	2 - 10 3 - 11	4 - 10 5 - 11	6 - 10 5 - 11	8 - 10 9 - 11					
6	2 - 4 3 - 11	5 - 11	4 - 6 5 - 11	4 - 8 9 - 11	4 - 10				
8	2 - 8 1 - 3	4 - 8 1 - 5	6 - 8 1 - 5	1 - 9	8 - 10 1 - 11	4 - 8 1 - 11			
9	X	X	X	X	X	X	X		
10	2 - 4 1 - 3	1 - 5	4 - 6 1 - 5	4 - 8 1 - 9	4 - 10 1 - 11	1 - 11	4 - 8	X	
11	1 - 3	2 - 4 1 - 5	2 - 6 1 - 5	2 - 8 1 - 9	2 - 10 1 - 11	2 - 4 1 - 11	2 - 8	X	2 - 4

	x	
q	0	1
1	2,0	3,0
2	4,0	5,0
3	6,0	5,0
4	8,0	9,0
5	10,0	11,0
6	4,0	11,0
8	8,0	1,0
9	10,1	1,0
10	4,0	1,0
11	2,0	1,0

Tabela de Implicação

1									
2 - 4 3 - 5	2								
2 - 6 3 - 5	4 - 6	3							
~~2 - 8 3 - 9~~	~~4 - 8 5 - 9~~	~~6 - 8 5 - 9~~	4						
2 - 10 3 - 11	4 - 10 5 - 11	6 - 10 5 - 11	~~8 - 10 9 - 11~~	5					
2 - 4 3 - 11	5 - 11	4 - 6 5 - 11	~~4 - 8 9 - 11~~	4 - 10	6				
2 - 8 1 - 3	4 - 8 1 - 5	6 - 8 1 - 5	~~1 - 9~~	8 - 10 1 - 11	4 - 8 1 - 11	8			
							9		
2 - 4 1 - 3	1 - 5	4 - 6 1 - 5	~~4 - 8 1 - 9~~	4 - 10 1 - 11	1 - 11	4 - 8		10	
1 - 3	2 - 4 1 - 5	2 - 6 1 - 5	~~2 - 8 1 - 9~~	2 - 10 1 - 11	2 - 4 1 - 11	2 - 8		2 - 4	11

	x	
q	0	1
1	2,0	3,0
2	4,0	5,0
3	6,0	5,0
4	8,0	9,0
5	10,0	11,0
6	4,0	11,0
8	8,0	1,0
9	10,1	1,0
10	4,0	1,0
11	2,0	1,0

Tabela de Implicação

	1	2	3	4	5	6	8	9	10
2	~~2 - 4~~ ~~3 - 5~~								
3	2 - 6 3 - 5	~~4 - 6~~							
4	~~2 - 8~~ ~~3 - 9~~	~~4 - 8~~ ~~5 - 9~~	~~6 - 8~~ ~~5 - 9~~						
5	2 - 10 3 - 11	~~4 - 10~~ ~~5 - 11~~	6 - 10 5 - 11	~~8 - 10~~ ~~9 - 11~~					
6	~~2 - 4~~ ~~3 - 11~~	5 - 11	~~4 - 6~~ ~~5 - 11~~	~~4 - 8~~ ~~9 - 11~~	~~4 - 10~~				
8	2 - 8 1 - 3	~~4 - 8~~ ~~1 - 5~~	6 - 8 1 - 5	~~1 - 9~~	8 - 10 1 - 11	~~4 - 8~~ ~~1 - 11~~			
9									
10	~~2 - 4~~ ~~1 - 3~~	1 - 5	~~4 - 6~~ ~~1 - 5~~	~~4 - 8~~ ~~1 - 9~~	~~4 - 10~~ ~~1 - 11~~	1 - 11	~~4 - 8~~		
11	1 - 3	~~2 - 4~~ ~~1 - 5~~	2 - 6 1 - 5	~~2 - 8~~ ~~1 - 9~~	2 - 10 1 - 11	~~2 - 4~~ ~~1 - 11~~	2 - 8		~~2 - 4~~

	x	
q	0	1
1	2,0	3,0
2	4,0	5,0
3	6,0	5,0
4	8,0	9,0
5	10,0	11,0
6	4,0	11,0
8	8,0	1,0
9	10,1	1,0
10	4,0	1,0
11	2,0	1,0

Tabela de Implicação

	1	2	3	4	5	6	8	9	10
2	~~2 - 4~~ ~~3 - 5~~								
3	2 - 6 3 - 5	~~4 - 6~~							
4	~~2 - 8~~ ~~3 - 9~~	~~4 - 8~~ ~~5 - 9~~	~~6 - 8~~ ~~5 - 9~~						
5	2 - 10 3 - 11	~~4 - 10~~ ~~5 - 11~~	6 - 10 5 - 11	~~8 - 10~~ ~~9 - 11~~					
6	~~2 - 4~~ ~~3 - 11~~	5 - 11	~~4 - 6~~ ~~5 - 11~~	~~4 - 8~~ ~~9 - 11~~	~~4 - 10~~				
8	~~2 - 8~~ ~~1 - 3~~	~~4 - 8~~ ~~1 - 5~~	~~6 - 8~~ ~~1 - 5~~	~~1 - 9~~	~~8 - 10~~ ~~1 - 11~~	~~4 - 8~~ ~~1 - 11~~			
9									
10	~~2 - 4~~ ~~1 - 3~~	1 - 5	~~4 - 6~~ ~~1 - 5~~	~~4 - 8~~ ~~1 - 9~~	~~4 - 10~~ ~~1 - 11~~	1 - 11	~~4 - 8~~		
11	1 - 3	~~2 - 4~~ ~~1 - 5~~	2 - 6 1 - 5	~~2 - 8~~ ~~1 - 9~~	2 - 10 1 - 11	~~2 - 4~~ ~~1 - 11~~	~~2 - 8~~		~~2 - 4~~

	x	
q	0	1
1	2,0	3,0
2	4,0	5,0
3	6,0	5,0
4	8,0	9,0
5	10,0	11,0
6	4,0	11,0
8	8,0	1,0
9	10,1	1,0
10	4,0	1,0
11	2,0	1,0

Tabela de Implicação

	1	2	3	4	5	6	8	9	10
2	2 - 4 3 - 5								
3	2 - 6 3 - 5	4 - 6							
4	2 - 8 3 - 9	4 - 8 5 - 9	6 - 8 5 - 9						
5	2 - 10 3 - 11	4 - 10 5 - 11	6 - 10 5 - 11	8 - 10 9 - 11					
6	2 - 4 3 - 11	5 - 11	4 - 6 5 - 11	4 - 8 9 - 11	4 - 10				
8	2 - 8 1 - 3	4 - 8 1 - 5	6 - 8 1 - 5	1 - 9	8 - 10 1 - 11	4 - 8 1 - 11			
9									
10	2 - 4 1 - 3	1 - 5	4 - 6 1 - 5	4 - 8 1 - 9	4 - 10 1 - 11	1 - 11	4 - 8		
11	1 - 3	2 - 4 1 - 5	2 - 6 1 - 5	2 - 8 1 - 9	2 - 10 1 - 11	2 - 4 1 - 11	2 - 8		2 - 4

11 (11)

10 (11) (10)

9 (11) (10) (9)
6
8 (11) (10) (9) (8)
5
6 (11) (6,10) (9) (8)

5 (5,11) (6,10) (9) (8)
3 3
4 (5,11) (6,10) (9) (8) (4)
2 2
3 (3,5,11) (6,10) (9) (8) (4)
1 1 1
2 (3,5,11) (2,6,10) (9) (8) (4)

1 (1,3,5,11) (2,6,10) (9) (8) (4)

Classes de equivalência: (1,3,5,11) (2,6,10) (9) (8) (4)

(1,3,5,11) - a
(2,6,10) - b
(9) - c
(8) - d
(4) - e

	x	
q	0	1
1	2,0	3,0
2	4,0	5,0
3	6,0	5,0
4	8,0	9,0
5	10,0	11,0
6	4,0	11,0
8	8,0	1,0
9	10,1	1,0
10	4,0	1,0
11	2,0	1,0

Tabela Reduzida (minimizada)

	x	
q	0	1
a	b,0	a,0
b	e,0	a,0
c	b,1	a,0
d	d,0	a,0
e	d,0	c,0

Minimize a tabela de estados abaixo

Present State q^{ν}	Input x^{ν}							
	0	1	2	3	0	1	2	3
1	6	2	1	1	0	0	0	0
2	6	3	1	1	0	0	0	0
3	6	9	4	1	0	0	1	0
4	5	6	7	8	1	0	1	0
5	5	9	7	1	1	0	1	0
6	6	6	1	1	0	0	0	0
7	5	10	7	1	1	0	1	0
8	6	2	1	8	0	0	0	0
9	9	9	1	1	0	0	0	0
10	6	11	1	1	0	0	0	0
11	6	9	4	1	0	0	1	0
	$q^{\nu+1}$				z^{ν}			

Resposta:

	$q^{\nu} \backslash x^{\nu}$	0	1	2	3
(4, 5)	a	$a, 1$	$b, 0$	$f, 1$	$c, 0$
(6, 9)	b	$b, 0$	$b, 0$	$c, 0$	$c, 0$
(1, 8)	c	$b, 0$	$d, 0$	$c, 0$	$c, 0$
(2)	d	$b, 0$	$e, 0$	$c, 0$	$c, 0$
(3)	e	$b, 0$	$b, 0$	$a, 1$	$c, 0$
(7)	f	$a, 1$	$d, 0$	$f, 1$	$c, 0$

$q^{\nu+1}, z^{\nu}$